PRACTITIONER KNOWLEDGE REPOSITORY
on Banking Regulation and Risk Management

MODULE 6

KVA, ColVA & Capital Costs

Completing the XVA Waterfall

The Final Two Floors

A building can stand without its top floors — but its true cost cannot be calculated without them. CVA, DVA, FVA, and MVA form the structure. KVA and ColVA are the final two floors: the rent on regulatory capital, and the value of collateral optionality. Without them, a bank cannot price derivatives at the true cost of doing business.

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Series: Basel I–IV → FRTB → CVA/DVA → FVA/MVA → ISDA, Netting & Collateral → KVA, ColVA & Capital Costs → Practitioner's Dictionary

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Section 1 — Setting the Scene: The Incomplete Waterfall

1.1 The XVA Waterfall to Date

The preceding four modules built the XVA framework from the ground up. Each adjustment addresses a specific cost embedded in an OTC derivative trade that is not captured by the risk-free valuation:

Module	Adjustment	What It Captures
M3	**CVA**	Expected loss from counterparty default — the credit cost of bilateral derivative exposure
M3	**DVA**	Expected gain from own default — the bilateral mirror of CVA
M4	**FVA**	Cost of funding uncollateralised derivative exposures on the dealer's balance sheet
M4	**MVA**	Cost of funding Initial Margin posted under UMR — the locked-up collateral cost
M6	**ColVA**	Value of the cheapest-to-deliver optionality in multi-currency CSAs — covered in this module
M6	**KVA**	Cost of regulatory capital consumed by the trade over its life — covered in this module

1.2 What Has Been Missing

The four adjustments covered in Modules 3 and 4 address credit risk and funding cost. They answer the question: what does it cost to fund and bear credit risk on this trade? But two further costs have been systematically underpriced or ignored entirely in pre-Basel III derivative markets:

- Capital cost: every derivative trade consumes regulatory capital. That capital has a cost — shareholders require a return on equity. If that cost is not priced into the trade, the bank is effectively subsidising the client.
- Collateral optionality: many CSAs allow counterparties to post collateral in multiple currencies or asset classes. The cheapest option changes over time as rates and spreads move. That optionality has a value — and one party in the trade is implicitly granting it to the other.

1.3 Why KVA and ColVA Matter Now

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Post-Basel III, regulatory capital is genuinely scarce. CET1 requirements of 10.5% (minimum plus conservation buffer), G-SIB surcharges of up to 3.5%, and stress capital buffers have made capital allocation a binding constraint for most global dealers. A trade that is profitable on a pre-capital basis can destroy shareholder value if its capital consumption is not priced in. KVA makes that cost explicit.

ColVA has grown in importance as multi-currency CSAs have proliferated, particularly in cross-border transactions. The cheapest-to-deliver option embedded in a EUR/USD/GBP-eligible CSA is not free. In low-rate environments with compressed cross-currency basis, it may be small. In stressed environments with wide basis and divergent central bank policies, it can be material.

1.4 How This Module Is Structured

This module follows a layered approach for both ColVA and KVA: plain English introduction first, conceptual framework with formula second, full derivation and worked example third. This structure is intentional — these adjustments are often dismissed as ‘quant topics’ by non-quantitative practitioners, but their commercial implications — pricing, capital allocation, client selection — affect everyone in a derivatives business.

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Section 2 — ColVA: Plain English Introduction

2.1 What Is ColVA in One Sentence

In plain English
ColVA is the value of the right to choose which currency or asset you post as collateral — and the cost to the counterparty of granting you that choice.

2.2 The Cheapest-to-Deliver Intuition

Imagine your CSA allows you to post collateral in either EUR cash or USD cash. Today, EUR OIS is 3.20% and USD OIS (SOFR) is 4.80%. If you post EUR cash, the receiver earns 3.20%. If you post USD cash, the receiver earns 4.80%. You will naturally post EUR cash — it costs you less to fund, and the difference in rates is money in your pocket.

Next quarter, central bank policy shifts. EUR OIS rises to 3.80%, USD OIS falls to 4.20%. The gap has narrowed. You might switch to posting USD. Next year, the gap reverses. You switch back.

This is the cheapest-to-deliver (CTD) option. You are always choosing whichever eligible collateral is cheapest for you to post. That flexibility has a value. The party receiving the collateral is implicitly granting you a free option — and ColVA is the price of that option.

ANALOGY

ColVA is like a loyalty programme where you can pay with whichever currency is weakest that day. The shop is always receiving less value than it would if you had to pay in a fixed currency. The discount the shop gives you is ColVA.

2.3 Why That Choice Has a Value — And Who Pays

In a single-currency CSA (EUR cash only), there is no choice — ColVA is zero by construction. In a multi-currency CSA, the choice exists and ColVA is positive. The party with the choice — the collateral poster — benefits. The party receiving collateral bears the cost.

In practice, the receiving party (typically the dealer) should charge for this optionality in the initial pricing of the trade. If it does not, it is giving away the option for free. Over a five-year trade with a volatile cross-currency basis, that cost can be significant.

2.4 ColVA vs FVA — Related but Distinct

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ColVA is frequently confused with FVA because both involve the cost of collateral. The distinction is important:

Dimension	FVA	ColVA
Source	Funding cost of uncollateralised exposure — the dealer funds the trade on its own balance sheet	Optionality value from choosing the cheapest eligible collateral in a multi-currency CSA
When non-zero	When there is no CSA or the trade is partially uncollateralised	When the CSA allows multiple eligible currencies or asset types
Driver	Dealer's own funding spread above OIS	Cross-currency basis and spread differentials between eligible collateral types
Can co-exist?	Yes — a trade can have both FVA (on uncollateralised portion) and ColVA (on collateralised portion with multi-currency CSA)	Yes — same trade

2.5 Who Cares About ColVA in Practice

ColVA is material for:

- Cross-currency swaps and long-dated FX derivatives — where CSAs are commonly multi-currency and tenors are long enough for basis to be volatile
- Trades between dealers with different home currencies — a EUR-base dealer trading with a USD-base dealer under a EUR/USD eligible CSA
- Any trade where the CSA eligible collateral schedule includes government bonds in multiple jurisdictions

ColVA is immaterial (effectively zero) for:

- Single-currency cash CSAs — the most common structure in standardised bilateral trading
- Centrally cleared trades — CCP margin rules specify narrow eligible collateral with no optionality
- Short-dated trades — where there is insufficient time for basis to move significantly

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Section 3 — ColVA: Full Derivation

3.1 The Multi-Currency CSA

A multi-currency CSA specifies a set of eligible collateral types — cash in multiple currencies, government bonds, or both. For each eligible asset, the CSA defines:

- The applicable rate paid on cash collateral (typically the OIS rate for that currency: ESTR for EUR, SOFR for USD, SONIA for GBP)
- Haircuts for non-cash collateral (see Module 5)
- Whether substitution is permitted and at what frequency

The collateral poster has the right, on each Valuation Date, to post whichever eligible asset minimises their funding cost. This right is the CTD option.

3.2 The CTD Option — Formal Definition

Let there be n eligible collateral currencies/assets indexed i = 1,...,n. For each, define r_i(t) as the OIS rate applicable to collateral of type i at time t. The collateral poster will choose at each point in time the asset with the lowest applicable rate:

```
r_CTD(t) = min{ r_1(t), r_2(t), ..., r_n(t) }
```

The dealer (collateral receiver) discounts future cash flows at the rate r_CTD(t) rather than a fixed OIS rate. The difference between discounting at a fixed rate and discounting at r_CTD(t) is the ColVA.

3.3 ColVA Formula — Full Derivation

Single Eligible Currency (Baseline)

With only one eligible currency (say EUR cash), r_CTD(t) = r_EUR(t) always. There is no optionality. The trade is discounted at EUR OIS throughout. ColVA = 0.

> **In plain English**
> If you can only pay in EUR, there is no cheapest-to-deliver choice. You always pay in EUR. No option, no ColVA.

Two-Currency Case — ColVA as the Value of a Spread Option

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With two eligible currencies (EUR and USD), the collateral poster will at each point post whichever is cheaper. The ColVA is the present value of the expected savings from this choice over the trade's life:

```
ColVA = integral_0^T N(t) * E[max(r_USD(t) - r_EUR(t), 0) *
1_{CTD=EUR} + max(r_EUR(t) - r_USD(t), 0) * 1_{CTD=USD}] * df(t) dt
```

This simplifies to the expected value of the spread option: the ColVA is the value of being able to post the lower-rate currency at each point. In closed form, assuming the spread s(t) = r_USD(t) − r_EUR(t) follows a mean-reverting process:

```
ColVA = integral_0^T N(t) * E[max(s(t), 0)] * df(t) dt
```

Where:

- N(t) = trade notional at time t (or EPE profile for a derivatives portfolio)
- s(t) = r_USD(t) − r_EUR(t) = the collateral spread between eligible currencies
- E[max(s(t), 0)] = the expected value of the positive spread — this is the value of a zero-strike call option on the spread
- df(t) = risk-free discount factor at time t

General Multi-Asset Case

With n eligible collateral types, the CTD option is a multi-asset exchange option. The ColVA is:

```
ColVA = integral_0^T N(t) * E[r_base(t) - min{r_1(t),...,r_n(t)}] *
                                df(t) dt
```

Where r_base(t) is the base collateral rate (the rate the dealer would receive without the multi-currency option, typically the OIS rate of the trade's pricing currency). The ColVA is always non-negative — adding more eligible currencies can only increase the value of the option, never decrease it.

3.4 Worked Example A — Single Currency CSA (EUR Cash Only)

Trade parameters (consistent with Module 4):

- Notional: EUR 100M
- Tenor: 5 years
- Trade type: Interest Rate Swap (pay fixed, receive floating)

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- Fixed rate: 3.20% (at-the-money)
- Eligible collateral: EUR cash only
- EUR OIS (ESTR): 3.20%

Since only EUR cash is eligible, the collateral poster has no choice. r_CTD(t) = r_EUR(t) = ESTR at all times. The trade is discounted at ESTR throughout.

```
ColVA = 0  (single eligible currency, no CTD optionality)
```

KEY INSIGHT

ColVA is zero by construction for any single-eligible-currency CSA, regardless of the level of rates. The zero does not mean collateral is free — FVA and MVA still apply if the trade is uncollateralised or requires IM. ColVA specifically measures the optionality value, which requires at least two eligible currencies to be non-zero.

3.5 Worked Example B — Multi-Currency CSA

Same trade, same parameters. CSA amended to allow EUR cash, USD cash, and GBP gilts (≤5Y maturity, 2% haircut).

Step 1: Identify Eligible Rates at Inception

Eligible Collateral	Applicable Rate	All-in Cost to Poster	CTD at Inception?
EUR Cash	ESTR = 3.20%	**3.20%**	**YES ✓**
USD Cash	SOFR = 4.80%	4.80%	No
GBP Gilts (≤5Y, 2% haircut)	SONIA = 4.20% – haircut carry	~4.05%	No

At inception, EUR cash is CTD. The spread to the next-cheapest option (USD cash) is 160 bps.

Step 2: Model the Spread Process

We model the EUR/USD collateral spread s(t) = r_USD(t) − r_EUR(t) as a mean-reverting (Ornstein-Uhlenbeck) process:

```
ds(t) = kappa * (theta - s(t)) dt + sigma_s * dW(t)
```

- kappa = mean reversion speed = 0.30 (spread reverts to long-run mean over ~3 years)
- theta = long-run mean spread = 1.20% (USD-EUR structural rate differential)
- sigma_s = spread volatility = 0.80% p.a. (cross-currency basis volatility)

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- s(0) = 1.60% (current spread at inception)

Step 3: Calculate ColVA via Monte Carlo

Running 10,000 paths of the spread process and computing the expected CTD saving at each quarterly time step over 5 years:

Year	E[s(t)] (%)	E[max(s(t),0)] (%)	Disc Factor	ColVA Contribution
1	1.45%	1.41%	0.969	EUR 1.37M
2	1.35%	1.32%	0.939	EUR 1.24M
3	1.28%	1.26%	0.910	EUR 1.15M
4	1.24%	1.22%	0.882	EUR 1.08M
5	1.22%	1.20%	0.855	EUR 1.02M
TOTAL ColVA				**EUR 5.86M**

ColVA = EUR 5.86M on a EUR 100M 5Y IRS. This represents approximately 5.86% of notional — material relative to the trade's bid-offer spread. A dealer that does not charge for this option is giving away EUR 5.86M of value.

Step 4: Sensitivity Analysis

Parameter Shift	ColVA	Change vs Base
Base case	**EUR 5.86M**	—
Spread vol +50% (σ_s = 1.20%)	**EUR 7.42M**	+EUR 1.56M (+27%)
Mean reversion faster (kappa = 0.60)	**EUR 4.91M**	−EUR 0.95M (−16%)
GBP also cheaper than EUR at inception	**EUR 6.88M**	+EUR 1.02M (+17%)
Single currency (EUR only)	**EUR 0**	−EUR 5.86M (−100%)

3.6 ColVA vs FVA — Interaction and Double-Count Risk

A common modelling error is to double-count ColVA and FVA. The risk arises because both involve the economics of collateral. The correct treatment:

- FVA applies to the uncollateralised component of the trade — the exposure above the threshold and MTA that is not covered by VM.

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- ColVA applies to the collateralised component — the portion where VM is posted but the poster has a choice of currency.
- For a fully collateralised trade with a single-currency CSA: FVA ≈ 0, ColVA = 0.
- For a fully collateralised trade with a multi-currency CSA: FVA ≈ 0, ColVA > 0.
- For a partially collateralised trade with a multi-currency CSA: both FVA and ColVA are non-zero, and apply to different portions of the exposure. They should not be added without ensuring they apply to mutually exclusive parts of the exposure.

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Section 4 — K(Kapital/ Capital)VA: Plain English Introduction

4.1 What Is KVA in One Sentence

In plain English (K as in Kapital and means Capital)
KVA is the annual rent a bank must charge on every derivative trade to cover the cost of the regulatory capital that trade requires it to hold — and to earn the return its shareholders demand on that capital.

4.2 Why Capital Is Not Free

Regulatory capital is the equity buffer banks are required to hold against potential losses. Under Basel III, the minimum requirement for a bank with a G-SIB surcharge is roughly 13–16% of Risk-Weighted Assets. This capital comes from shareholders — people who invested in the bank expecting a return.

A bank's shareholders do not accept the risk-free rate on their investment. They could earn that by buying government bonds. They invest in a bank because they expect a higher return — typically 10–15% per annum on equity. This is the hurdle rate.

In plain English
Think of capital like a security deposit on a rented flat. Your landlord (the regulator) requires you to lock up some cash before you can trade. That locked-up cash earns nothing. KVA is the annual charge to recover the cost of having that cash tied up instead of deployed elsewhere at the hurdle rate.

4.3 The Intuition

Every derivative trade consumes capital because it creates risk-weighted assets (RWA). More RWA means more required capital. More required capital means more shareholder equity tied up in the trade. That equity has an opportunity cost — the hurdle rate. KVA is the present value of those opportunity costs over the trade's life.

If a bank prices a trade without including KVA, it is effectively saying: "our shareholders' capital can be used for free." Over time, this destroys shareholder value. Trades that look profitable on a pre-KVA basis may be value-destroying on a post-KVA basis.

4.4 KVA Before Basel III — Why It Was Ignored

Before Basel III, regulatory capital requirements were lower, less risk-sensitive, and more easily optimised. The pre-2008 world had:

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- Basel I/II capital requirements that significantly understated derivative CCR
- Internal models (IMM) that allowed sophisticated banks to reduce capital to very low levels
- No leverage ratio constraint — gross balance sheet could grow without capital impact
- Low hurdle rates (8–10% ROE) reflecting lower systemic risk perceptions

In this environment, the capital cost of a derivative trade was small enough to be buried in the bid-offer spread. Banks did not need to make it explicit.

4.5 KVA After Basel III — Capital Scarcity Changes Everything

Post-Basel III, four changes made capital costs impossible to ignore:

1. SA-CCR replaced IMM as the standard for CCR capital, increasing RWA for many derivatives books
2. CET1 requirements rose to 10.5% minimum plus G-SIB surcharge of up to 3.5% plus stress capital buffer of up to 2.5%
3. The leverage ratio (3% minimum, 5% for G-SIBs) added a non-risk-sensitive constraint that bites for low-RWA high-notional trades
4. Shareholder ROE expectations reset higher (12–15%) as investors demanded compensation for higher perceived systemic risk

MAGNITUDE CHECK	*For a EUR 100M 5Y IRS with a BBB counterparty, the annual capital charge under Basel III (CET1 10.5% + G-SIB 1.5% = 12% CET1, hurdle rate 12%) is approximately EUR 180K–220K per year. Over five years, discounted, this is EUR 700K–900K — material relative to the trade's economics.*

4.6 Who Pays KVA

In a dealer-to-client trade, KVA is ultimately borne by the client through wider bid-offer spreads or explicit pricing adjustments. In competitive markets, pressure to compress spreads may lead dealers to absorb part of the KVA cost themselves — effectively subsidising client trades. Over time, this produces a steady drag on shareholder returns.

In dealer-to-dealer markets, both sides are KVA-aware. The KVA cost is priced bilaterally, and trades that are capital-inefficient for both sides may be replaced by cleared equivalents where the capital treatment is more favourable.

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Section 5 — KVA: Conceptual Framework with Formula

5.1 The Building Blocks

KVA is constructed from three inputs, each feeding the next:

Step	Input	Explanation
1	**EAD (Exposure at Default)**	Calculated under SA-CCR. The regulatory measure of how much the bank is exposed to if the counterparty defaults today. For an IRS: EAD = alpha × (RC + PFE), where alpha = 1.4.
2	**RWA (Risk-Weighted Assets)**	RWA = EAD × Risk Weight. Risk weight depends on counterparty credit rating: BBB corporate = 100%, A = 50%, AAA/AA = 20%. RWA determines how much capital the bank must hold.
3	**K(t) (Required Capital)**	K(t) = RWA(t) × CET1 requirement. This is the actual equity capital the bank must hold at time t. K(t) varies over the trade's life as the EAD profile evolves.
4	**KVA**	KVA = present value of [hurdle rate × K(t)] over the trade's life. This is the cost of holding that capital, discounted to today.

5.2 The KVA Formula

```
KVA = h * integral_0^T E[K(t)] * df(t) dt
```

Where:

- h = hurdle rate (required return on equity, typically 10–15% p.a.)
- E[K(t)] = expected required capital at time t, averaged over scenarios
- df(t) = risk-free discount factor at time t
- T = trade maturity

In plain English
Read this as: "For each year of the trade's life, multiply the capital the trade requires by the hurdle rate. That gives the annual cost. Add up all those annual costs, discounted to today. That total is KVA."

5.3 The Hurdle Rate

The hurdle rate h is the bank's required return on equity. It reflects:

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- The risk-free rate (government bond yield) — the minimum acceptable return
- An equity risk premium — compensation for the additional risk of holding bank equity vs government bonds
- A bank-specific premium — reflecting the bank's own credit risk, leverage, and business model volatility

Bank Type	Typical Hurdle Rate	Rationale
G-SIB (top tier)	**12–15%**	Higher capital requirements; investor expects premium for systemic risk exposure
Large regional bank	**10–12%**	Lower capital requirements; smaller systemic footprint
Worked examples (this module)	**12%**	Conservative G-SIB assumption. Consistent with Module 4 funding spread assumptions.

5.4 E[K(t)] — Expected Capital Over the Trade's Life

The required capital K(t) is not constant over the trade's life. For an IRS:

- The Replacement Cost (RC) component of SA-CCR EAD reflects current MtM — this fluctuates as rates move
- The PFE component reflects the potential future exposure — this follows a humped profile, peaking at roughly one-third of tenor for an IRS and declining as the trade approaches maturity
- Therefore K(t) follows a similar humped profile for an IRS, peaking in years 1–2 and declining thereafter

In practice, E[K(t)] is computed by Monte Carlo simulation of interest rate scenarios, calculating the SA-CCR EAD → RWA → K at each path and time step, and averaging.

5.5 The Time Integral — Why Future Capital Matters

The integral in the KVA formula captures the full lifetime capital cost. A five-year trade that consumes EUR 5M of capital in year 3 has a present value KVA contribution from that year of:

```
KVA_year3 = h * K_year3 * df(3) = 12% * EUR 5M * 0.910 = EUR 546K
```

The integral sums these contributions across all years. For long-dated trades, the out-year contributions are significant even after discounting, making KVA particularly sensitive to tenor.

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Section 6 — KVA: Full Derivation & Worked Example

6.1 SA-CCR Inputs for KVA

The SA-CCR framework (covered in Module 3) provides the EAD used in KVA. For a vanilla IRS netting set with no CSA (uncollateralised, for maximum KVA illustration):

```
EAD = alpha * (RC + PFE)  where alpha = 1.4
```

```
RC = max(V - C, 0)  where V = MtM, C = collateral held
PFE = multiplier * AddOn_IRS
AddOn_IRS = SF * delta * d * MF * Notional
```

For a par IRS at inception (V = 0, no collateral, C = 0):

- RC = max(0 − 0, 0) = 0
- Supervisory Factor (SF) for IR: 0.50%
- Delta (δ): +1 for receive-fixed (payer is −1)
- Effective Notional (d): EUR 100M × modified duration ≈ EUR 100M × 4.5 = EUR 450M (duration-adjusted for 5Y IRS)
- Maturity Factor (MF) for unmargined: min(sqrt(M/1), sqrt(1)) = 1.0 for M ≥ 1 year

```
AddOn_IRS = 0.50% * 1 * EUR 450M * 1.0 = EUR 2.25M
PFE = 1.0 * EUR 2.25M = EUR 2.25M  (multiplier = 1.0 for RC = 0)
EAD = 1.4 * (0 + EUR 2.25M) = EUR 3.15M
```

6.2 RWA and Capital Requirement

BBB corporate counterparty under the Standardised Approach:

```
RWA = EAD * Risk Weight = EUR 3.15M * 100% = EUR 3.15M
CET1 Minimum (10.5%) = EUR 3.15M * 10.5% = EUR 330.75K
```

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6.3 Stress Capital Buffer

The Stress Capital Buffer (SCB) replaced the static capital conservation buffer under Basel III final rules (effective 2020 in the US, phased in globally). The SCB is bank-specific, calibrated to the stressed losses projected in the annual supervisory stress test. For most major dealers, the effective SCB is 2.5–5.0%.

Capital Layer	Rate	Purpose
CET1 Minimum	**4.5%**	Absolute minimum; breach triggers restrictions on distributions
Capital Conservation Buffer	**2.5%**	Static buffer; replaced by SCB under US Basel III Endgame
Stress Capital Buffer (SCB)	**2.5–5.0%**	Bank-specific stress buffer from supervisory stress test; minimum 2.5%
G-SIB Surcharge	**1.0–3.5%**	Additional CET1 requirement for Global Systemically Important Banks
Worked example total (Bucket 2 G-SIB)	**12.0%**	4.5% + 2.5% SCB + 3.5% (Bucket 3 used for illustration) + 1.5% G-SIB

6.4 G-SIB Surcharge — Structure and KVA Impact

The G-SIB framework assigns banks to buckets (1–5) based on systemic importance scores across five categories: size, interconnectedness, substitutability, complexity, and cross-jurisdictional activity. Each bucket carries an additional CET1 requirement:

Bucket	G-SIB Surcharge	Example Banks	KVA Impact
1	**1.0%**	Various EU banks	Modest KVA uplift — marginal capital cost increase
2	**1.5%**	BNP Paribas, Deutsche Bank	Meaningful uplift — KVA approximately 14% higher vs non-G-SIB
3	**2.0%**	Barclays, Goldman, Morgan Stanley	KVA approximately 19% higher vs non-G-SIB
4	**2.5%**	Citi, BofA, Wells Fargo	KVA approximately 24% higher vs non-G-SIB
5 (empty)	**3.5%**	Buffer bucket — currently vacant	Regulatory deterrent against further systemically important concentration

COMPETITIVE IMPLICATION *A Bucket 3 G-SIB (2% surcharge) competing against a non-G-SIB on the same trade has a structurally higher KVA. For capital-intensive trades, this creates a genuine competitive*

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disadvantage. One response: G-SIBs favour cleared products (lower CCR capital) and push bilateral trades toward compression and netting to reduce RWA.

6.5 KVA Worked Example — EUR 100M 5Y IRS

Using the Module 4 trade parameters: EUR 100M notional, 5Y IRS, BBB counterparty, no CSA (uncollateralised). Bank is a Bucket 2 G-SIB with 12% total CET1 requirement and 12% hurdle rate.

Step 1: SA-CCR EAD Profile Over Trade Life

The PFE profile for a 5Y IRS peaks around year 1.5–2 and declines as the trade approaches maturity. The following approximates the expected EAD at each year-end:

Year	Remaining Tenor	Adj. Notional	AddOn	EAD (alpha=1.4)	Disc Factor
1	4Y	EUR 400M	EUR 2.00M	EUR 2.80M	0.969
2	3Y	EUR 330M	EUR 1.65M	EUR 2.31M	0.939
3	2Y	EUR 250M	EUR 1.25M	EUR 1.75M	0.910
4	1Y	EUR 150M	EUR 0.75M	EUR 1.05M	0.882
5	0Y	EUR 0	EUR 0	EUR 0	0.855

Step 2: Capital Requirement at Each Year-End

Year	EAD	RWA (100%)	K(t) = RWA*12%	Annual KVA = h*K(t)*df(t)
1	EUR 2.80M	EUR 2.80M	EUR 336K	12% * EUR 336K * 0.969 = EUR 39.1K
2	EUR 2.31M	EUR 2.31M	EUR 277K	12% * EUR 277K * 0.939 = EUR 31.2K
3	EUR 1.75M	EUR 1.75M	EUR 210K	12% * EUR 210K * 0.910 = EUR 22.9K
4	EUR 1.05M	EUR 1.05M	EUR 126K	12% * EUR 126K * 0.882 = EUR 13.3K
5	EUR 0	EUR 0	EUR 0	EUR 0
KVA TOTAL				**EUR 106.5K**

RESULT

KVA = EUR 106.5K on a EUR 100M 5Y IRS with a BBB counterparty, no CSA, Bucket 2 G-SIB, 12% hurdle rate. This is approximately 1.07 bps of notional — a small but non-trivial cost that must be recovered in the bid-offer spread or explicit pricing.

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Step 3: Sensitivity Analysis

Scenario	KVA	vs Base
Base case (BBB, no CSA, 12% hurdle, Bucket 2)	**EUR 106.5K**	—
Hurdle rate 15%	**EUR 133.1K**	+EUR 26.6K (+25%)
A-rated counterparty (RW 50%)	**EUR 53.3K**	−EUR 53.2K (−50%)
Bucket 4 G-SIB (14.5% total CET1)	**EUR 128.7K**	+EUR 22.2K (+21%)
10Y tenor (vs 5Y)	**EUR 248.3K**	+EUR 141.8K (+133%)
Fully collateralised (daily VM, zero TH)	**EUR 38.2K**	−EUR 68.3K (−64%) — collateral reduces EAD significantly

6.6 Adjoint Algorithmic Differentiation (AAD) for KVA Sensitivities

Why Standard Monte Carlo Greeks Are Prohibitively Expensive for KVA

Computing KVA requires simulating the expected regulatory capital profile E[K(t)] across thousands of Monte Carlo paths and hundreds of future time steps. Each path involves repricing the full netting set at every time point under SA-CCR. The resulting computation is already expensive. The problem compounds dramatically when sensitivities are required.

Standard finite difference (bump-and-revalue) Greeks require re-running the full Monte Carlo simulation for each risk factor independently. A trading book with N risk factors requires N+1 full simulations — one base run plus one bumped run per factor. For a realistic interest rate book with 500+ tenor/currency combinations, this is computationally intractable for intraday or real-time KVA Greeks. The cost scales as O(N) in the number of risk factors.

KVA sensitivities that are commercially essential but computationally expensive under finite difference include: dKVA/dRate (interest rate delta across the yield curve), dKVA/dCredit Spread (counterparty credit sensitivity), dKVA/dVol (vega — relevant for swaption books), and dKVA/dNotional (trade-level capital attribution). Without these Greeks, the XVA desk cannot hedge KVA risk or allocate capital costs to individual trades at granular level.

What AAD Is — Plain English

Adjoint Algorithmic Differentiation (AAD) is a technique from computational mathematics that computes the gradient of a function with respect to all its inputs simultaneously, in a single backward pass through the computation graph — at approximately 2–5 times the cost of a single forward evaluation, regardless of the number of input risk factors.

The intuition: the forward pass of a Monte Carlo simulation is a sequence of computations — simulate market states, reprice trades, aggregate exposure, compute capital. AAD augments this forward pass by recording the computational graph (a “tape”) of every intermediate operation. A single backward pass — the adjoint sweep — then propagates derivatives from the output (KVA) back through the tape to every input (each market risk factor) simultaneously. The result is the complete gradient vector in one pass.

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AAD analogy: imagine computing your route from A to B (forward pass), then simultaneously identifying every road junction that affected your journey time — not by redriving every alternative route, but by replaying the single journey in reverse and recording sensitivities at each junction. AAD does this for every risk factor in the KVA computation in one backward sweep.

Forward Mode vs Adjoint Mode — Why Adjoint Wins for XVA

Algorithmic differentiation has two modes. Forward mode computes the derivative of the output with respect to one input per pass — cost scales as O(N) in the number of inputs, identical to finite difference in computational complexity. Adjoint (reverse) mode computes the derivative of one output with respect to all inputs in a single backward pass — cost scales as O(1) relative to the number of inputs, subject to a constant factor of 2–5x the forward pass cost.

For KVA and XVA Greeks, the computation has one output (KVA) and many inputs (N risk factors). This is precisely the regime where adjoint mode dominates: one output, many inputs — the adjoint sweep delivers all N sensitivities at ~3–5x the cost of a single Monte Carlo run, compared to N+1 runs under finite difference. For N = 500 risk factors, the speed-up is approximately 100–150x.

KVA Greeks Enabled by AAD

The following KVA sensitivities become computationally tractable under AAD but are prohibitive under finite difference at production scale:

- dKVA/dRate (IR Delta): sensitivity of KVA to each point on the yield curve. Required for interest rate hedging of the capital profile. A 5Y IRS book may have 30–50 curve points per currency, each requiring a separate bump under finite difference.
- dKVA/dCredit Spread (CS01): sensitivity of KVA to counterparty credit spread. As counterparty PD increases, EAD-weighted capital cost changes. Critical for understanding how KVA evolves as a counterparty deteriorates.
- dKVA/dVol (Vega): sensitivity of KVA to implied volatility. Relevant for swaption and exotic books where PFE — and therefore SA-CCR EAD — is volatility-sensitive. AAD propagates vol sensitivities through the full repricing engine.
- dKVA/dNotional (Trade-Level Attribution): the marginal KVA contribution of each trade to the netting set. Essential for trade-level capital allocation and pricing new trades at the margin. Under finite difference, this requires one full re-simulation per trade in the book.

Computational Cost Comparison

Illustrative comparison for KVA Greek calculation on a 500-trade IRS book with 300 risk factors:

- Finite difference: 301 full Monte Carlo simulations required (1 base + 300 bumped). At 30 seconds per run: ~2.5 hours total. Intraday Greeks: not feasible.
- AAD: 1 forward pass + 1 adjoint sweep = ~3–5x single run cost. At 30 seconds forward: ~90–150 seconds total for all 300 Greeks simultaneously. Speed-up: ~60–100x. Intraday Greeks: fully feasible.
- Real-time KVA at trade execution (sub-second): requires GPU-accelerated AAD or neural network surrogate models trained on AAD-generated Greeks. Active research area; production at select G-SIBs for vanilla products.

Industry Adoption and Reference Implementations

- CompatibL: specialist XVA and AAD platform widely deployed at G-SIBs. Provides AAD-enabled CVA, FVA, and KVA Greeks as production infrastructure. The CompatibL framework is the most

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cited AAD implementation in practitioner literature (Risk.net, Global Derivatives conference proceedings).

- Quaternion / MSCI: risk analytics platform with integrated AAD for XVA and FRTB CVA sensitivities. Used for both economic XVA Greeks and regulatory SA-CVA sensitivities under the FRTB CVA framework.
- Numerix: cross-asset analytics with AAD support used as pricing engine input to XVA systems. Particularly relevant for exotic product books where repricing is expensive.
- In-house: JPMorgan, Goldman Sachs, Barclays, and UBS have all published or presented internal AAD implementations for XVA Greeks. The build vs buy decision typically favours buy for smaller XVA books and build for G-SIBs with bespoke capital models.

ColVA Greeks use the same AAD infrastructure at negligible marginal cost: dColVA/dBasis (cross-currency basis delta), dColVA/dSpreadVol (vega of the CTD spread option), and dColVA/dCorrelation are all delivered in the same backward sweep as KVA Greeks.

For developers building KVA systems: the standard reference is Savine (2018) “Modern Computational Finance: AAD and Parallel Simulations” (Wiley), which provides worked C++ implementations directly applicable to XVA Monte Carlo engines. The CompatibL open-source AAD library is the practitioner starting point.

6.7 KVA and the Leverage Ratio

The Basel III leverage ratio (3% minimum, 3.5–5% for G-SIBs) provides a non-risk-sensitive backstop. For derivative trades with low RWA but large gross notional, the leverage ratio can bind before the RWA-based capital requirement. When the leverage ratio binds, the effective capital cost is calculated on gross exposure rather than risk-weighted exposure — potentially making the KVA significantly higher.

This is particularly relevant for:

- FX forwards and short-dated derivatives where RWA is low but notional is large
- Highly rated counterparties (AA/AAA) where the standardised risk weight of 20% produces very low RWA
- Netting sets where netting benefit is large — net RWA is low but gross leverage exposure remains high

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Section 7 — The Complete XVA Waterfall

7.1 All Six Adjustments in One Framework

The complete XVA framework for an OTC derivative trade now encompasses six valuation adjustments, each addressing a specific cost not captured by risk-free pricing:

Order	XVA	Economic Meaning	Primary Driver	Module
1	**CVA**	Expected loss from counterparty default	Counterparty CDS spread, EPE profile	M3
2	**DVA**	Expected gain from own default	Own CDS spread, ENE profile	M3
3	**FVA**	Funding cost of uncollateralised exposure	Dealer funding spread, EPE/ENE	M4
4	**MVA**	Cost of funding Initial Margin	SIMM IM amount, funding spread	M4
5	**ColVA**	Value of CTD collateral optionality	Cross-currency basis, spread vol	M6
6	**KVA**	Cost of regulatory capital over trade life	SA-CCR EAD, RWA, hurdle rate	M6

7.2 Interactions and Double-Count Risks

The six XVA components are not independent. Several interact and potentially double-count if not handled carefully:

CVA vs FVA

A defaultable counterparty has a lower effective funding cost for the dealer — the probability of default reduces the expected funding burden. This creates a tension: charging both full CVA and full FVA on the same exposure double-counts the credit risk component. In practice, FVA is typically computed on the risk-neutral expectation of exposure (ignoring credit events), while CVA handles the default scenario separately.

FVA vs ColVA

FVA applies to uncollateralised exposure; ColVA applies to collateralised exposure where multiple currencies are eligible. They apply to mutually exclusive parts of the exposure for a given trade. However,

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a trade with a threshold CSA (partially collateralised) has both: FVA on the exposure above the threshold, ColVA on the collateralised portion.

MVA vs KVA

Posting IM reduces CCR capital (lower EAD under SA-CCR) and therefore reduces KVA. But IM increases MVA (funding cost of the locked-up collateral). The net effect depends on the relative size of the capital cost vs the funding cost. For most trades, MVA > KVA reduction from IM, meaning IM posting increases the total XVA burden — which is why dealers seek to minimise IM through portfolio compression and clearing optimisation.

7.3 Three Collateral Regime Case Studies

Case 1: Uncollateralised (No CSA)

Trade: EUR 100M 5Y IRS, BBB counterparty, no CSA, Bucket 2 G-SIB, 12% hurdle rate, 80 bps funding spread.

XVA	Amount	Driver
CVA	EUR 245K	Full bilateral credit exposure; no collateral offsetting EPE
DVA	−EUR 42K	Own credit benefit on ENE (typically smaller than CVA)
FVA	EUR 312K	Full EPE funded at 80 bps spread — largest component
MVA	EUR 0	No CSA — no IM requirement (below UMR threshold assumed)
ColVA	EUR 0	No CSA — no collateral optionality
KVA	EUR 107K	Full SA-CCR EAD; no collateral reducing capital
TOTAL XVA	**EUR 622K**	62.2 bps of notional — significant cost of an uncollateralised trade

Case 2: Fully Collateralised (Daily VM, EUR Cash Only, Zero Threshold)

XVA	Amount	Driver
CVA	~EUR 8K	Near-zero — daily VM settles exposure; residual from MPoR only
DVA	−EUR 2K	Near-zero correspondingly
FVA	~EUR 12K	Near-zero — funded by VM flows, not dealer balance sheet
MVA	EUR 88K	SIMM IM required; locked-up funding cost at 80 bps spread
ColVA	EUR 0	Single eligible currency (EUR) — no CTD optionality
KVA	EUR 38K	Collateral reduces SA-CCR EAD significantly — lower capital
TOTAL XVA	**EUR 144K**	14.4 bps — 77% reduction vs uncollateralised. MVA now dominant.

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Case 3: Partially Collateralised (Threshold EUR 2M, Multi-Currency CSA)

XVA	Amount	Driver
CVA	EUR 82K	Partial exposure above EUR 2M threshold uncollateralised
DVA	−EUR 14K	Bilateral mirror
FVA	EUR 105K	Funding cost on uncollateralised portion above threshold
MVA	EUR 88K	SIMM IM on full portfolio (IM not affected by threshold)
ColVA	**EUR 5.86M**	Multi-currency CSA (EUR/USD/GBP) — full CTD optionality value
KVA	EUR 71K	Partial collateralisation reduces EAD vs uncollateralised
TOTAL XVA	**EUR 6.19M**	ColVA dominates. Multi-currency CSA transforms the XVA profile.

KEY INSIGHT

The three-case comparison shows that the dominant XVA component shifts radically with the collateral regime. Uncollateralised: FVA dominates. Fully collateralised (single currency): MVA dominates. Partially collateralised with multi-currency CSA: ColVA dominates — often by a wide margin. Collateral regime selection is one of the most consequential pricing decisions in derivative trading.

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Section 8 — XVA Desk P&L Attribution

8.1 How XVA P&L Is Generated

XVA P&L arises from two sources:

- Mark-to-market (MtM) P&L: the change in the fair value of XVA reserves as market inputs move — credit spreads, interest rates, funding curves, cross-currency basis. This P&L is unrealised and flows through the income statement under IFRS 13.
- Realised P&L: cash flows associated with XVA hedging (CDS premiums for CVA hedges, repo costs for FVA funding, custody fees for IM) and any default events that crystallise CVA losses.

The XVA desk holds a reserve — the total XVA charge on the book — against which it runs hedging strategies. P&L attribution tracks how each component of the reserve changed and why.

8.2 The XVA Desk Structure

Large dealers operate a centralised XVA desk that is the single internal counterparty for all XVA charges. Trading desks book trades at risk-free (clean) prices; the XVA desk charges the XVA costs back to the trading desk via internal transfer pricing. This structure:

- Centralises credit, funding, and capital risk in one place, enabling portfolio-level netting and hedging
- Provides clean P&L attribution: trading desk sees risk-free P&L; XVA desk sees XVA P&L
- Allows the XVA desk to hedge CVA with CDS, fund positions via the Treasury desk, and manage capital allocation centrally

8.3 P&L Attribution by Component

CVA P&L

P&L Driver	Explanation
Credit spread moves	Widening counterparty CDS spreads increase CVA reserve (negative P&L for CVA desk). Tightening reduces reserve (positive P&L).
EPE profile changes	Rate moves change the EPE profile of derivative positions. A large rate move that increases EPE will increase CVA even if credit spreads are unchanged.
New trades	Each new trade adds incremental CVA. The XVA desk charges this to the trading desk as a day-one cost. The charge reduces over the trade's life as exposure decays.

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Default events	A counterparty default crystallises the CVA loss — the reserve is released against the actual loss. If CVA was correctly sized, the P&L impact is zero. If undersized, a residual loss hits the P&L.
Hedge P&L	CDS premiums paid on CVA hedges are a cash cost. CDS MtM moves offset CVA reserve moves when hedges are well-matched.

FVA P&L

P&L Driver	Explanation
Funding spread moves	Dealer's own funding spread widening increases FVA reserve (negative P&L). Tightening reduces it. This is the primary source of FVA P&L volatility.
Collateral regime changes	A counterparty agreeing to post VM converts an uncollateralised exposure to a near-zero FVA exposure — a large one-time FVA P&L release.
Rate moves	Large rate moves change the EPE/ENE profile, changing the size of the uncollateralised exposure that must be funded.

MVA P&L

P&L Driver	Explanation
SIMM recalibration	Annual SIMM updates change risk weights and correlations. A recalibration that increases IR delta weights will increase SIMM IM, increasing MVA reserve.
Sensitivity moves	Rate moves change the delta and vega sensitivities of the portfolio, directly changing SIMM IM and therefore MVA.
Funding spread moves	Same as FVA — MVA scales linearly with the funding spread. A 10 bps funding spread tightening reduces MVA by approximately 10 bps on the notional IM amount.

ColVA P&L

P&L Driver	Explanation
Collateral spread moves	A widening of the EUR/USD basis (USD rates rising relative to EUR) increases the value of the EUR CTD option, increasing ColVA reserve (negative P&L for the receiver).
CTD switches	When the cheapest currency switches (e.g. from EUR to USD as rates converge), the ColVA profile changes — potentially large one-time P&L if the switch is unexpected.
Spread volatility changes	Higher cross-currency basis volatility increases the value of the CTD option, increasing ColVA. Compressed volatility reduces it.

KVA P&L

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P&L Driver	Explanation
RWA changes	Rate moves that increase EAD increase RWA and therefore the KVA reserve. Netting improvements reduce RWA and release KVA reserve.
Regulatory updates	SA-CCR parameter changes, G-SIB bucket reassignments, or SCB updates directly change the capital requirement and therefore KVA. Basel IV Endgame output floor changes will be a significant KVA P&L event for many banks.
Hurdle rate revisions	A change in the bank's required return on equity (e.g. from 12% to 14% following an investor relations update) re-prices the entire KVA book.

8.4 Hedging the XVA Book

Not all XVA components are equally hedgeable:

XVA	Hedgeability	Typical Hedge	Residual Risk
CVA	Good — CDS market liquid for IG names	Single-name or index CDS on counterparty	Wrong-way risk; basis between CDS and actual exposure
DVA	Poor — hedging own default is complex and controversial	Own bond buybacks; contingent capital instruments	Accounting inconsistency; regulatory restrictions on self-hedging
FVA	Moderate — funding spread partially hedgeable	Repo, own bond issuance, internal Treasury funding curves	Basis between hedge instruments and actual funding cost
MVA	Moderate — IM amount can be predicted; funding cost hedgeable	Repo of IM collateral; forward starting FX for currency mismatch	SIMM recalibration risk; collateral substitution optionality
ColVA	Limited — cross-currency basis partially hedgeable	Cross-currency basis swaps; FX options on spread	CTD switching is path-dependent; difficult to hedge precisely
KVA	Very limited — regulatory capital risk not directly hedgeable	Portfolio compression; clearing optimisation; capital efficient structuring	Regulatory change risk is unhedgeable; hurdle rate changes are unhedgeable

8.5 Accounting Treatment — IFRS 13

Under IFRS 13, derivative fair values must include all relevant valuation adjustments that a market participant would consider. CVA and DVA are explicitly required. FVA, MVA, ColVA, and KVA are included by most major dealers on the basis that they represent costs that would be reflected in an exit price in an orderly market transaction.

The practical implication: changes in CVA, DVA, FVA, MVA, ColVA, and KVA all flow through the P&L of derivatives held at fair value through profit or loss. This creates P&L volatility driven by market inputs that

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are not directly related to the economics of the trading book — credit spread volatility, funding spread volatility, and cross-currency basis volatility.

8.6 Worked Case: Full XVA P&L Decomposition

EUR 100M 5Y IRS, BBB counterparty, multi-currency CSA (EUR/USD, threshold EUR 2M), Bucket 2 G-SIB. One year has elapsed since inception. Market moves:

- Counterparty CDS spread: widened 25 bps
- Dealer funding spread: unchanged
- EUR/USD basis: widened 20 bps (USD rates rose relative to EUR)
- Rates: +50 bps parallel shift (EPE profile unchanged by assumption)

XVA	P&L	Driver	Sign
CVA	**−EUR 28K**	Counterparty CDS +25 bps × EPE profile	Loss — wider credit spread increases reserve
DVA	**+EUR 4K**	DVA moves partially offset CVA	Gain (partially offsets CVA loss)
FVA	−EUR 2K	Funding spread unchanged; minor decay of FVA reserve	Negligible
MVA	**+EUR 6K**	Time decay of IM — one year of IM funding cost earned	Gain — earning carry on IM position
ColVA	**−EUR 82K**	EUR/USD basis +20 bps increases CTD option value	Loss — CTD optionality increased (receiver's cost increased)
KVA	**+EUR 11K**	Time decay of KVA reserve — one year of capital cost earned	Gain — earning carry on capital position
NET XVA P&L	**−EUR 91K**	ColVA and CVA losses dominate; MVA/KVA carry partially offsets	Net loss — driven by basis widening and credit spread widening

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Section 9 — Regulatory & Strategic Dimensions

9.1 Basel IV / Endgame — Output Floor Impact on KVA

Basel IV (the 'Basel III Endgame' in US nomenclature) introduces an output floor: internal model outputs for RWA cannot be less than 72.5% of the Standardised Approach RWA. For banks that have historically used internal models (IMM) to reduce CCR capital significantly below SA-CCR levels, the output floor will increase effective RWA — and therefore KVA — substantially.

IMPACT ESTIMATE	*Industry estimates suggest the output floor will increase RWA for CCR by 15–40% for sophisticated dealers currently using IMM. This translates directly into higher KVA charges on derivative books. Trades that were previously KVA-neutral or KVA-positive may become KVA-negative post-Basel IV.*

The strategic response for most G-SIBs: accelerate portfolio compression, increase clearing rates, and reprice bilateral trades to reflect the higher capital cost. KVA will become an increasingly explicit line item in client pricing as Basel IV is implemented (2025–2028 across jurisdictions).

9.2 FRTB CVA Framework

The FRTB CVA framework (covered in Module 2) imposes capital charges on the CVA desk's hedging activities. This interacts with KVA:

- CVA hedges (CDS positions) that reduce CVA volatility now carry their own capital charge under FRTB-CVA
- This makes CVA hedging more expensive, potentially leading desks to carry more unhedged CVA risk
- Unhedged CVA creates larger CVA P&L volatility, which increases the effective KVA on the CVA book itself
- The combined effect: FRTB-CVA increases the all-in cost of maintaining a hedged CVA book, making the economics of the XVA business more capital-intensive

9.3 Strategic Implications

KVA changes the strategic calculus of derivative businesses in three dimensions:

Product Pricing

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Long-dated, uncollateralised, or capital-intensive trades must carry a KVA premium. Trades that appear competitive at clean prices may be uneconomic once KVA is included. Dealers that explicitly price KVA lose volume to dealers that do not — but dealers that do not price KVA destroy shareholder value over time.

Client Selection

BBB counterparties consume 5x the capital of AAA/AA counterparties (100% vs 20% risk weight). KVA creates an incentive to allocate scarce derivative capacity to highly-rated clients. Low-rated clients face either wider bid-offer spreads or reduced access to derivative markets from capital-constrained dealers.

Balance Sheet Optimisation

KVA incentivises three structural responses:

- Compression: netting down offsetting positions reduces gross EAD without changing economic risk
- Clearing: CCP-cleared trades attract lower CCR capital (5-day MPoR, CCP risk weight 2%) vs bilateral (10-day MPoR, full counterparty risk weight)
- Novation and restructuring: breaking long-dated bilateral trades into shorter-dated cleared equivalents reduces peak KVA

9.4 The Future of XVA

The XVA framework is not static. Three developments are reshaping it:

- AI/ML in XVA computation: Monte Carlo simulation of 10,000+ paths for CVA, FVA, MVA, ColVA, and KVA is computationally intensive. ML-based approximations (Gaussian process regression, neural network surrogates) are being deployed by leading dealers to enable real-time XVA pricing during trade negotiation rather than overnight batch.
- Real-time KVA: the ability to compute KVA in real-time changes the capital allocation conversation — relationship managers can see the capital cost of a proposed trade before agreeing terms, enabling capital-efficient deal structuring.
- Regulatory evolution: Basel IV output floors, potential updates to SA-CCR, and evolving G-SIB methodology will continue to shift the KVA landscape. Dealers with flexible, model-driven KVA infrastructure will adapt faster than those with hardcoded capital assumptions.

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Glossary — New Terms from Module 6

The following terms are introduced in Module 6 and will be carried into Module D (the Practitioner's Dictionary) with full three-layer entries.

Term	Definition	Module
ColVA	Collateral Valuation Adjustment. The value of the cheapest-to-deliver optionality in a multi-currency CSA. Zero for single-eligible-currency CSAs; positive and potentially material for multi-currency agreements.	M6
KVA	Capital Valuation Adjustment. The present value of the cost of regulatory capital consumed by a derivative trade over its life. KVA = h * integral_0^T E[K(t)] * df(t) dt.	M6
Hurdle Rate	Required return on equity capital, typically 10–15% p.a. for major dealers. The rate at which capital cost is charged in KVA calculation.	M6
CTD Option	Cheapest-to-Deliver option. The right, under a multi-currency CSA, to post whichever eligible collateral type is cheapest at each Valuation Date. The source of ColVA.	M6
Collateral Spread	The differential between OIS rates applicable to different eligible collateral types. The primary driver of ColVA. Modelled as a mean-reverting stochastic process.	M6
Output Floor	Basel IV requirement that RWA calculated using internal models cannot fall below 72.5% of the Standardised Approach RWA. Increases KVA for banks currently using IMM.	M6
Stress Capital Buffer	Bank-specific CET1 buffer calibrated to supervisory stress test losses. Minimum 2.5%. Replaced the static capital conservation buffer under US Basel III Endgame.	M6
G-SIB Surcharge	Additional CET1 requirement (1.0–3.5%) for Global Systemically Important Banks, assigned by bucket based on systemic importance score. Increases effective KVA.	M6
XVA Desk	Centralised desk responsible for pricing, managing, and hedging all XVA adjustments. Acts as internal counterparty to trading desks, charging XVA costs via internal transfer pricing.	M6
Portfolio Compression	Netting down offsetting bilateral positions to reduce gross EAD without changing net economic risk. Reduces RWA and KVA. Facilitated by TriOptima and similar platforms.	M6
FRTB-CVA	The FRTB CVA capital framework imposing market risk charges on the CVA hedging book. Increases the capital cost of running a hedged CVA desk.	M6

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M3	CVA, DVA & Counterparty Credit Risk
M4	FVA & MVA — Funding & Margin Valuation Adjustments
M5	ISDA, Netting & Collateral
M6	**KVA, ColVA & Capital Costs — Completing the XVA Waterfall ✓ THIS MODULE**
MD	**Practitioner's Dictionary — XVA & Banking Regulation — NEXT**